

Cooperative Emergence in Multi-Agent Stag-Hunt

A Comparison of A2C, PPO and MAPPO

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Experimental Framework and Environments

3 Environments

- **Hunt** — joint stag capture
passive exploit vs. active
coordination
- **Harvest** — harvest mature
plants
temporal synchronization
- **Escalation** — sustain a streak
long-horizon commitment, partial
observability

3 Algorithms

- **A2C** $T = 5$ steps
- **PPO** $T = 256$ steps
- **MAPPO** $T = 256$,
centralized critic

Framework

- Independent Learners
- Shared weights
(flip_obs=True)
- Separate buffers
- 3 training variants:
standard, robust,
adaptive (Hunt)

Performance Comparison

No single algorithm wins everywhere

	Hunt (std, stag=T)	Harvest	Escalation
1 st	A2C 2394	A2C 1743	PPO 66.68
2 nd	PPO 344*	PPO 815	MAPPO 10.68
3 rd	MAPPO -22	MAPPO 654	A2C 0.26 [†]
Random	-1630	357	2.33
Key factor	Passive exploit	Update frequency	Rollout horizon

*PPO $\sigma=3324$: high variance, partial convergence. [†]Below random baseline.

Curriculum (adaptive): MAPPO 644 > PPO 634 > A2C 576 — all converge to stable foraging.

Why Update Mechanisms Matter

Harvest (Time Precision)

Mature plants degrade rapidly (prob. 0.1/step), opening a narrow 10-step cooperation window.

Frequent updates ($T=5$) capture the precise step triggering the cooperative reward immediately.

Long rollouts ($T=256$) dilute this temporal signal across hundreds of unrelated transitions.

Escalation (Commitment)

Sustaining a streak requires long-term commitment, but the counter is omitted from observations.

PPO's horizon ($T=256$) spans the entire interaction, ensuring correct credit assignment.

A2C ($T=5$) fails to link actions to future outcomes, rapidly collapsing into active defection.

Hunt (Exploit Resilience)

When the prey tracks the agents, remaining stationary emerges as a highly effective passive exploit.

Step-level updates ($T=5$) reinforce this behavior instantly, achieving a stable 2394 average reward.

Mixed trajectories introduce heavy noise into PPO's GAE, causing high instability ($\sigma=3324$).

Note: Domain randomization proves catastrophic for batch-based methods (MAPPO policy collapses to -4985), whereas step-level adjustments provide crucial resilience against environment non-stationarity.